Controlled Packet Capture and 802.11 Protocol Analysis Lab - Retries and Pseudo Headers

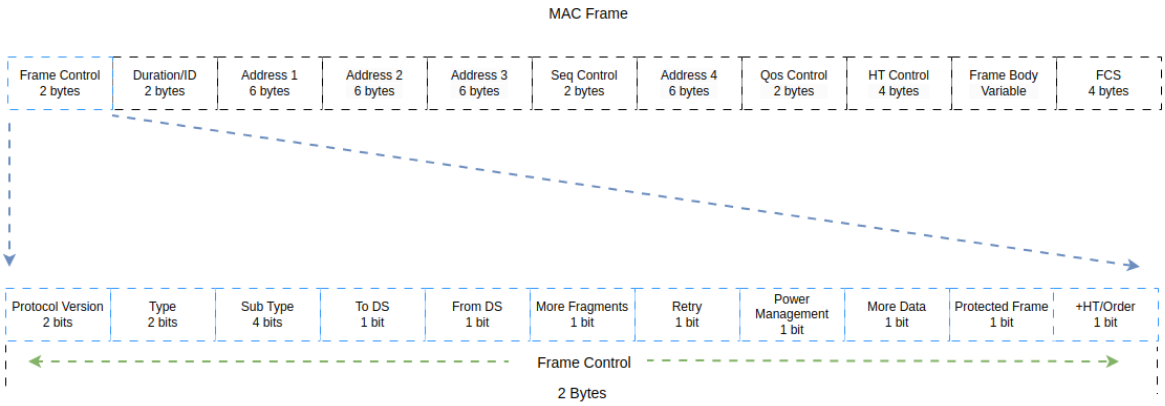
While troubleshooting using packet captures, it is essential to understand where to look to help identify possible issues. In this lab we will have a look at the Retry field and how we can use this information to isolate potential problems. Next we will have a look at what Pseudo Headers are and how this metadata can assist in our troubleshooting efforts.

Retry Field

So what is the "Retry Field" and "How can we identify and use" this information? Great Question! Simply put, a retry is when a Sending Station does not receive an "ACK / Acknowledgement" back from that transmission. Recall the Distributed Coordination Function process? A sending station needs to receive an "ACK" or the contention window doubles (up to a maximum) and the station tries again. The lack of reception of an "ACK" can come from different scenarios. Keep in mind, only unicast frames are acknowledged. Broadcast and multicast frames, such as beacons, are never ACKed, so they are never retried.

Ok, we now know what a retry is, where can we find this information? Another great question! Recall our previous conversations relating to MAC frames? Specifically, Frame Control? The 'Retry Field' is set in Frame Control.

MAC Frame Reference



Notice "Retry" field directly after "More Fragments"?

If you will be spending any amount of time analyzing wireless packet-capture, I suggest memorizing the MAC frame format. This will be time well spent!

Wireshark and Retry Field

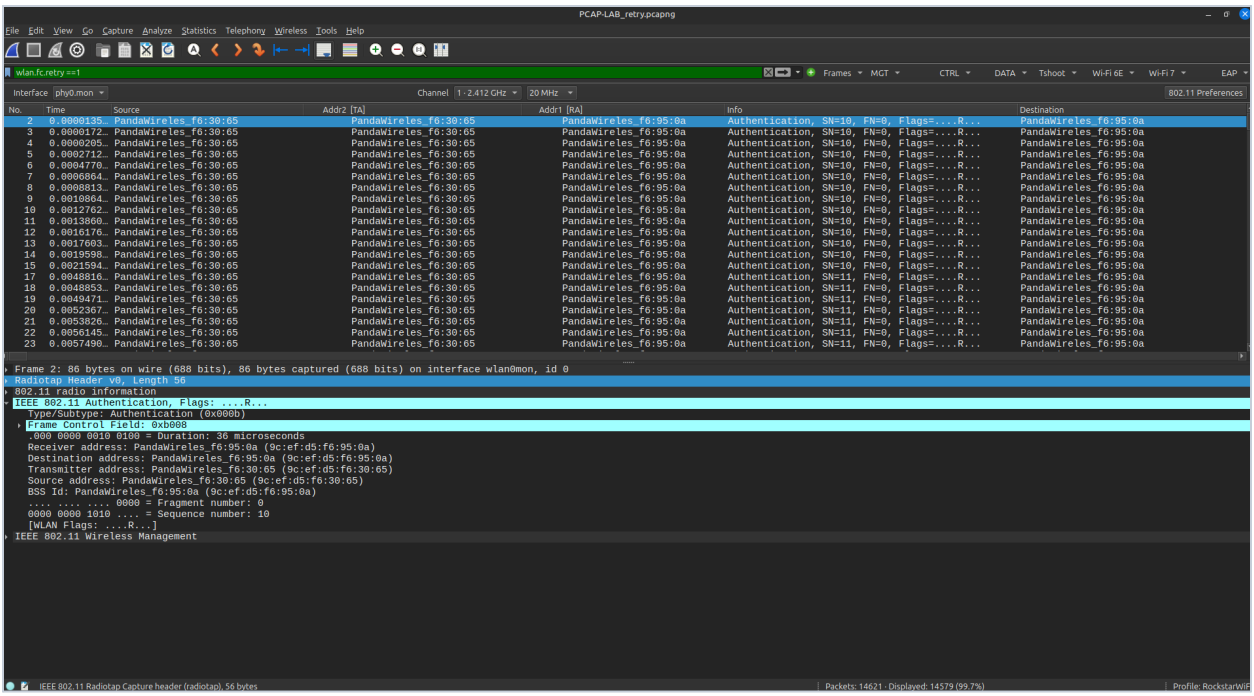
Let's start by looking at a packet-capture and applying a display filter relating to the "Retry Field".

Wireshark Display Filter - Retry

Notice how the filter calls out 'fc' (Frame Control)? We are calling out the Retry field in Frame Control! We passed the "1" option stating that we want to display all frames with the Retry set to "1" indicating that this is a retry frame.

```
wlan.fc.retry ==1
```

Result



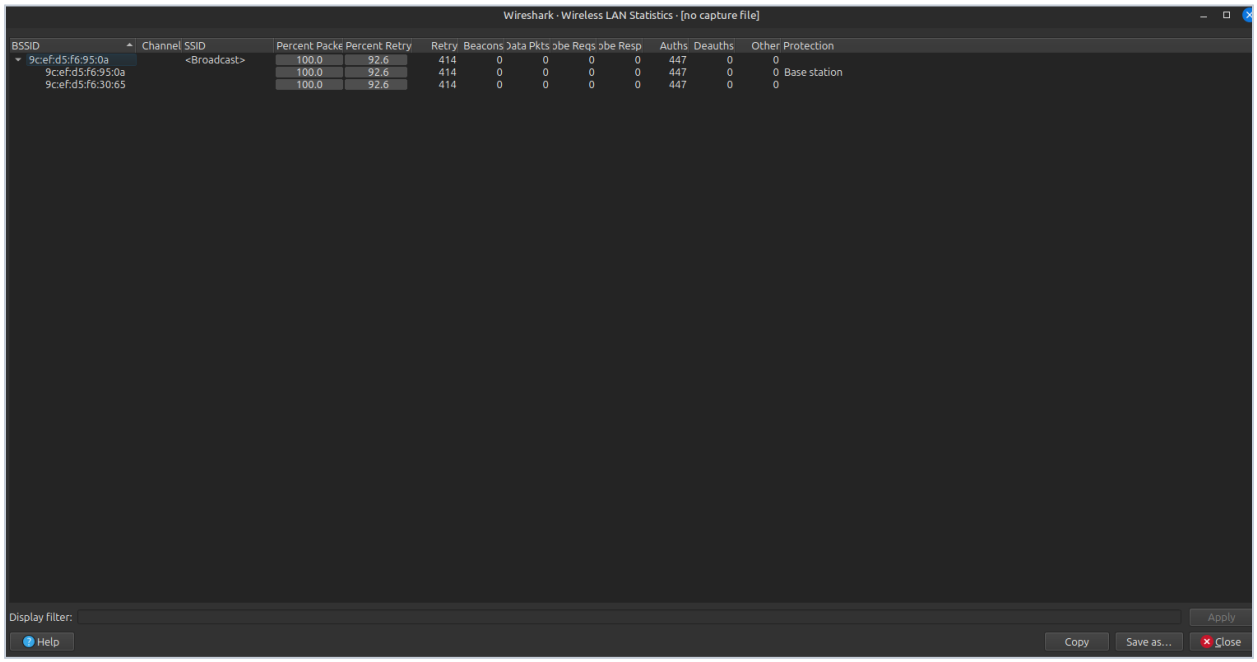
We now have our packet-capture filtered based on the Retry Field, which is located in the Frame Control field of the MAC frame. At this point there are a few ways we can gather valuable information.

First

Have a look at the bottom of the Wireshark window. We can see "Packets". We also see "Displayed". This will indicate how much, in percentage is displayed of the packet-capture based on the display filter. Since we are filtering on "Retry", the percentage will relate to Retries.

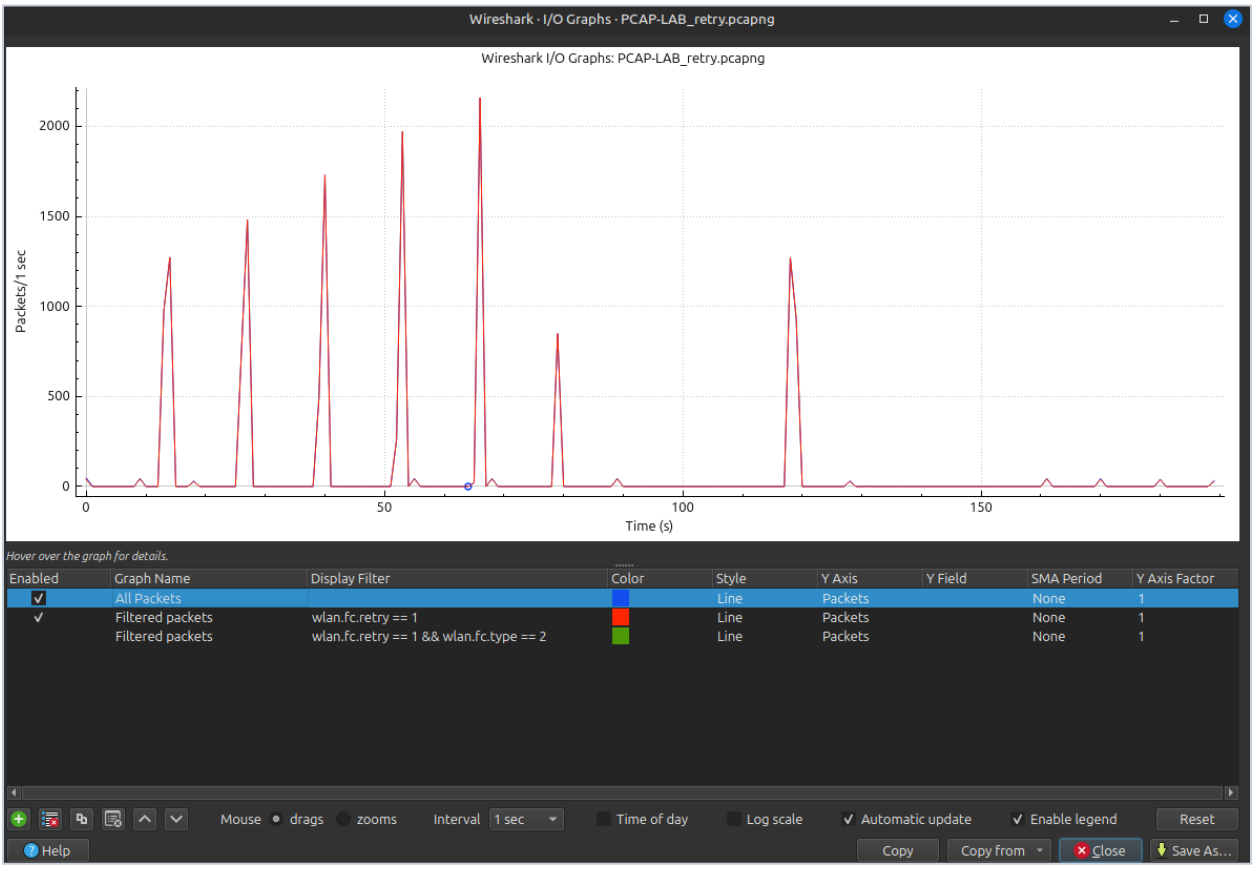
Second

Located on the top menu of Wireshark, there is an option called "Wireless". Select "Wireless" then "WLAN Traffic". This display will show the Wireless LAN traffic relating to the current packet-capture. By expanding the BSSID of our Lab Access-point, we will find valuable information. Notice the Retry Column?



Third

Using Wireshark "I/O Graphs" we can apply display filters and compare and contrast traffic patterns. This view is helpful in displaying a clean visual for understanding and reporting. To access the "I/O Graph", on the top menu, select, "Statistics", then "I/O Graphs".



We can see below our graph, we can add and apply filters as needed. There are other options available as well. I encourage everyone to use the I/O Graph display as it is very useful.

Notice, "wlan.fc.retry == 1" is the red line?

Retry Meaning

Great! We know where the Retry Field lives, how it is set, and how to identify it. But what does the "Retry Field" tell us?

This is a great question. The short answer is that it tells us, at the core, that the sender is not receiving an Acknowledgment. But why? That can take many forms and is variable to each environment. Here is a brief list of possible scenarios why the sender is not receiving an "ACK."

This is not an all-inclusive list. Please explore each situation independently.

- **RF interference** - Co-channel or adjacent-channel interference, or non-Wi-Fi noise such as microwaves and Bluetooth, corrupts frames so the receiver never ACKs them.

- **Weak signal or low SNR** - The receiver can't decode the frame cleanly. This is common at the edge of coverage or when a client sits far from the AP.
- **Collisions** - Two stations transmit at the same time, which happens more often on busy channels or with hidden nodes (stations that can't hear each other but can both reach the AP).
- **Data rate mismatch** - The sender uses a rate too high for the current conditions, and rate adaptation hasn't stepped down yet.
- **Receiver problems** - The client is busy or overloaded, or has a poor antenna or driver, so it misses the frame or the ACK.
- **Lost ACK** - The data frame arrived fine, but the ACK was lost on the way back, so the sender retries a frame that was already received.
- **Roaming or mobility** - A client moving between APs can have frames retried while the connection is degrading.

To finalize, a retry in a packet-capture is evidence of a problem and not proof of a problem. Keep in mind, in a wireless environment there will always be a certain level of retries. The level at which retries become impactful depends on many variables. To that end, there are guidelines indicating at which level, or more specifically, which percentage of retries, will have a negative impact. I encourage exploration and application to the environment you are analyzing.

Pseudo Headers

So what is a Pseudo Header?

Simply put, a pseudo header (Radiotap) is metadata added by the wireless capture interface to provide additional information about the received 802.11 frame, such as RSSI, channel, and data rate. It is not transmitted over the air but provides useful context during wireless packet analysis.

Here are a few examples of the type of information that is provided.

- RSSI - Received signal strength (e.g., -55 dBm).
- Channel - Operating frequency (e.g., 5180 MHz / Channel 36).
- Data Rate - Transmission rate (e.g., 54 Mbps).
- MCS - Modulation and Coding Scheme.
- Antenna - Receiving antenna information.
- Channel Width - 20, 40, 80, or 160 MHz.

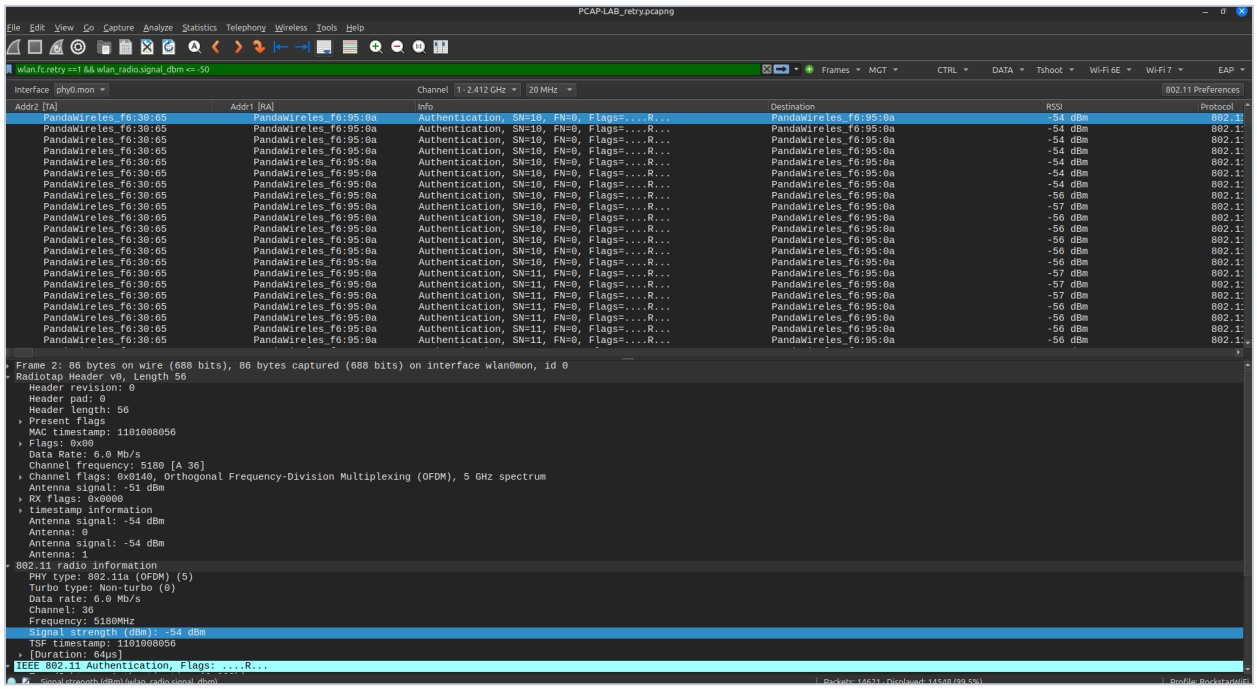
Great! Let's have a look at how we can use this information to assist with our troubleshooting efforts! Let's apply a Wireshark display filter to our retry packet-capture.

Let's have a look at the "Signal Strength" or "RSSI"!

```
wlan.fc.retry ==1 && wlan_radio.signal_dbm <= -70
```

We tell Wireshark to display all frames with retry set to 1 and signal strength less than or equal to (-70).

Wireshark Retry with Signal Strength Filter



Notice the "Signal Strength" located under "802.11 radio information"? This is useful information while troubleshooting. In our lab, the client is close to the AP (around -55 dBm), so the screenshot uses -50 to show the fields. In a real environment, -70 dBm is a better starting point for weak-signal retries.

Conclusion

We have explored the Retry Field, where it can be found in the Frame Control Field, and how it plays a role in troubleshooting via packet analysis. We then discussed pseudo headers, what they are, how to filter for signal strength, and potential ways to utilize this metadata during troubleshooting. It is important to understand that pseudo header information is never sent over the air. Pseudo header information is added by the driver of the capture adapter to provide Layer 1 information to augment Layer 2 information provided by 802.11 MAC frames.